Diffusion Models as Classifiers for EEG Seizure Detection

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1. Motivation & Core Question

- **Problem.** Epilepsy affects ∼1% of people. Seizure detection in EEG is still done *by hand* - slow, error-prone, hours of multi-channel data scanned window by window.
- **Gap.** *Discriminative* CNNs/RNNs dominate; *generative* models serve mostly for augmentation - rarely as the classifier itself, especially for biomedical *time series* [3].
- **Why generative?** A class-conditional model of the *signal* gives a likelihood + per-window uncertainty, not just a label - so a clinician can trust or defer each call.

Core question

One frozen conditional diffusion model, four variants - differing only in *how* its outputs become a decision: from *training-free* (V1, V2) to a *learned read-out* (V3, V4).

Which mechanism detects seizures most robustly?

2. A Diffusion Model is Secretly a Classifier

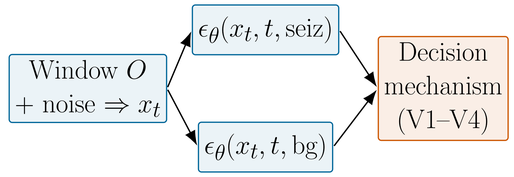
A **DDPM** learns to reverse a noising process; the forward step is

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon.$$

- Schedule $\bar{\alpha}_t$: 1 (signal x_0) $\to$ 0 (pure noise).
- Network $\epsilon_\theta(x_t, t, c)$ predicts the noise, *conditioned* on class label c.
- **Key idea** (Li et al. [1]): pick the class whose conditioning best explains the observed noise.
- In its purest form (V1, V2) - no extra classifier, no extra training: the generative model *is* the classifier.

3. How a Window Gets Classified

For an EEG window O: add noise, then score *both* class labels on the *same* noised input.



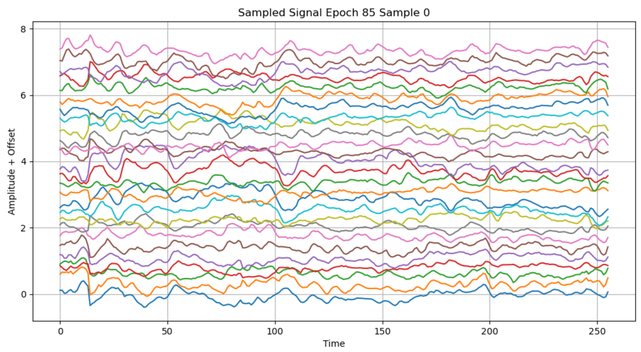
- Repeat over random timesteps + noise (e.g. N=50×) (V1), then aggregate per-draw scores into one decision.

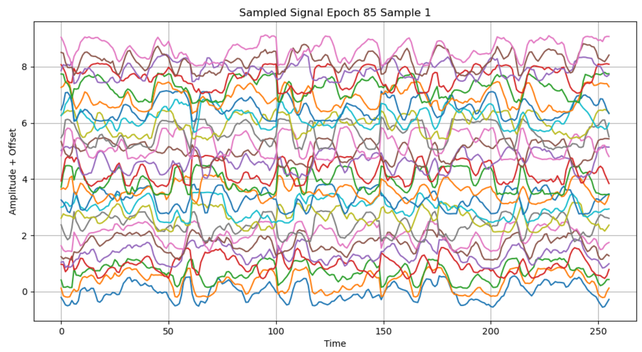
Data & Experimental Setup

- **Dataset.** TUSZ v2.0.6 - largest public corpus of clinical EEG with expert seizure annotations.
- **Input.** 27-channel scalp EEG, 2 s windows.
- **Task.** Binary per-window: *seizure* vs. *background*.
- **Protocol.** Official Temple University split - subject-disjoint (no patient in two splits).
- **Challenge.** Seizures <5% per split - *severe class imbalance*. Shapes training (few seizure examples) and evaluation (report sensitivity at fixed FA-rate, not only ROC-AUC / Balanced Acc.).

Generated EEG Signals - Conditional Samples from the Diffusion Model

Two class-conditional windows (2 seconds, 128 Hz) drawn from the trained model (epoch 85) $\epsilon_\theta(\cdot, \cdot, c)$ - 27 channels stacked with vertical offset.
Left: Background; **Right:** Seizure





V1 - Monte-Carlo Majority Vote [1, 3]

Idea. Which class label lets the denoiser better predict this window's noise?

Decision. Per draw, vote for class c with smaller error $e^{(c)} = \|\epsilon_\theta(x_t, t, c) - \epsilon\|_2^2$; *majority* over N draws.

Trade-off. Training-free, strong baseline - but coarse, poorly-calibrated confidence; metric choice (MAE/MSE) open.

V2 - Probability-Flow Log-Likelihood [2]

Idea. Don't count votes - measure how *probable* the window is per class.

Decision. Probability-flow ODE gives a *numerical* log-likelihood per class; decide by the ratio $\Delta\text{LL} = \log p_\theta(O|\text{seiz}) - \log p_\theta(O|\text{bg})$.

Trade-off. Optimal if densities are correct (Neyman–Pearson), best-calibrated score - but an ODE solve per window is costly.

V3 - Learned Critic [5]

Idea. Don't hand-pick the metric - let a small model *learn* what separates the classes.

Decision. Feed diffusion outputs (predicted noise / per-step errors) to a learned critic (CNN or gradient-boosted) outputting $P(\text{seiz} \mid O)$.

Trade-off. Most flexible ⇒ highest accuracy potential - but needs own training and can overfit the model's artifacts.

V4 - Siamese / Triplet Metric [4]

Idea. Learn a *distance* where a window sits closer to its own class's generated version.

Decision. Encoder trained with triplet loss + hard-negative mining; classify by smaller *average learned distance* over N runs.

Trade-off. Most expressive, gives continuous per-window uncertainty - but hardest to train: hard-negative mining is unstable.

References

[1] Li et al., ICCV 2023 · [2] Song et al., ICLR 2021 · [3] Vetter et al., Patterns 2024 · [4] Schroff et al., CVPR 2015 · [5] Kim et al., ICML 2023.

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